

Quality drift after the SKILL / TypeScript migration

A controlled artifact-chain review of the same UMSS feature request produced by the old and new Overmind runtimes.

EXECUTIVE VERDICT

No overall negative quality drift was found. The new runtime produces moderately and consistently better final and intermediate artifacts—about +10 quality points overall. The only clear regression is weaker semantic dependency review, which unnecessarily serializes frontend scaffold readiness behind backend delivery.

FINAL ARTIFACTS

92

NEW +10 POINTS

INTERMEDIATE

90

NEW +9 POINTS

OVERALL

91

NEW +10 POINTS

What changed for the better

- Cleaner separation of Telegram identity registration, OFFCHAIN_POINTS account creation, account lookup, counts, and frontend delivery.
- More concrete endpoint, error-envelope, time-window, recovery, and no-write semantics.
- Stronger traceability with less repetition: matched intermediate artifacts use 13% fewer words while carrying more actionable detail.
- Broader business clarification resolves ten material questions in the new run versus five in the old run.

INTERPRETATION

This is a strong case-study result, not a pure causal A/B test. The raw feature story is identical, but clarification answers and project blueprints differ. The evidence supports the new end-to-end process; it does not attribute the entire gain to TypeScript alone.

01 Method and comparison basis

The two repositories contain the same feature_requirements.txt content. Only capture metadata and feature identifiers differ in user_br_input.md. Quality was scored against five dimensions: source fidelity (30), precision and testability (20), cross-artifact traceability (20), implementation readiness and ordering (20), and auditability / signal-to-noise (10). Scores are comparative judgments with approximately ±3 points uncertainty.

FAIR PLAN BASELINE

The old implementation plan was assessed at master commit 184b403, immediately after semantic review. The current old plan contains later implementation checkmarks and follow-up steps; comparing that evolved plan with the unexecuted new plan would unfairly favor the old run.

02 Final artifact quality

Artifact	Old	New	Difference
Requirements EARS	84	92	New +8
Implementation plan	79	91	New +12
Combined final quality	82	92	New +10

Requirements EARS — why the new result is stronger

- Separates identity registration from the subsequent account-creation request instead of treating both as one onboarding transaction.
- Defines the last-24-hour window using request-arrival server time and defines manual refresh as visible recovery behavior.
- Distinguishes the mandatory Telegram user ID from optional profile fields and keeps invalid-input handling no-write.
- Makes duplicate account creation a deterministic business rejection while preserving automated verification for every major behavior.

Implementation plan — why the new result is stronger

- Uses concrete endpoint identities throughout the plan.
- Separates identity registration, account creation, and authoritative lookup into executable steps.
- Freezes all four REST resources and shared error-envelope semantics before consumer binding.
- Explicitly tests duplicate no-write behavior, zero-state counts, server-time window calculation, and non-sensitive failures.

03 Intermediate artifact quality by phase

Phase	Old	New	Winner
BR capture and clarification	76	90	New +14
EARS semantic review	88	92	New +4
Surface maps	81	89	New +8
Technical requirements	81	90	New +9
Contract delta	78	92	New +14
Implementation slices	77	91	New +14
Prerequisite gaps	76	92	New +16
Plan semantic review	88	81	Old +7

DENSITY IMPROVEMENT

Across matched intermediate artifacts, the new run uses 11,439 words versus 13,174 in the old run—a 13% reduction—while adding concrete endpoint, contract, error, timing, and recovery information.

Largest improvements

- BR clarification: ten material questions cover timing, visibility, validity, recovery, history, reliability, observability, integration, and rollout. The old run asks five.
- Contract delta: the new artifact introduces explicit lifecycle, count, and error-envelope deltas instead of one broad onboarding contract.
- Implementation slices: account creation is no longer hidden inside identity onboarding, and concrete endpoints anchor each slice.
- Prerequisite gaps: one canonical surface catalog replaces repeated per-requirement prerequisite blocks.

The one phase that regressed

Both semantic reviews detect missing backend and frontend repository readiness. The old review rates these blockers High and keeps frontend readiness independent. The new review rates them Medium and makes frontend scaffold readiness depend on completed backend count delivery, losing safe parallelism without a real contract, payload, schema, state, or repository prerequisite.

04 Why quality moved

- Broader clarification coverage resolves more business decisions before EARS generation.
- The new common-contract baseline explicitly models a shared error envelope; the old baseline records error shape as an unresolved uncertainty.
- New artifact rules preserve capability boundaries and concrete user-reachable surfaces more consistently.
- Technical artifacts bind to specific endpoints and label repository conclusions as inference when repositories are deferred.
- TypeScript gates improve deterministic structure, token coverage, source binding, and failure reporting.

WHAT TYPESCRIPT DID NOT SOLVE

Structural validation can prove that a dependency references an existing earlier step, but it cannot prove that the dependency is necessary. The new frontend-readiness dependency therefore passes validation even though it reduces executable parallelism. Semantic review remains model-judgment work and needs a stronger quality contract.

05 Residual quality risks

- Cross-artifact reconciliation: the new BR summary still says exactly one ACTIVE existing account, while EARS review correctly removes the ACTIVE qualifier. The final requirement is right, but upstream remains inconsistent.
- Clarification auditability: the new missing-data ledger preserves questions and target fields but not the verbatim user answers that produced the final values.
- Coordination semantics: new readiness steps omit the optional Coordination marker, so their machine-readable classification is weaker than their titles and bullets.
- Causal isolation: blueprint versions, selected stacks, clarification answers, and contract baselines differ, so a single paired run cannot isolate the tool migration effect.

06 Recommended actions

Priority	Action	Expected effect
P0	Remove the Step 1.7 → Step 1.6 dependency; keep page delivery dependent on backend count delivery.	Restore safe frontend scaffold parallelism.
P0	Add a semantic-review check that every dependency includes an evidence-backed necessity rationale.	Prevent valid-but-unnecessary serialization.
P1	Add an explicit reconciliation outcome when EARS review corrects behavior sourced from the BR summary.	Eliminate upstream/downstream contradictions.
P1	Persist concise verbatim clarification answers alongside field mappings.	Improve provenance and auditability.
P2	Run a frozen-input benchmark across multiple features and models.	Isolate SKILL/TS effects from human and blueprint differences.

07 Most important conclusions

- The migration is successful: approximately +10 quality points overall, with the largest gains in contract modeling, slicing, and prerequisite traceability.
- The new artifacts are denser and more executable, not merely more verbose or more formally structured.
- Clarification breadth appears to be the largest quality lever; preserve it independently of implementation technology.
- The main remaining risk has shifted from structural completeness to semantic optimization: dependency necessity, parallelism, and upstream reconciliation.
- A controlled multi-feature benchmark should be the next evaluation step before treating the observed gain as entirely caused by SKILL/TypeScript architecture.

FINAL ASSESSMENT

Keep the SKILL/TypeScript architecture. There is no evidence of negative migration drift in this feature, and there is material evidence of positive quality movement. Focus the next iteration on semantic dependency review and artifact reconciliation rather than expanding structural validation alone.